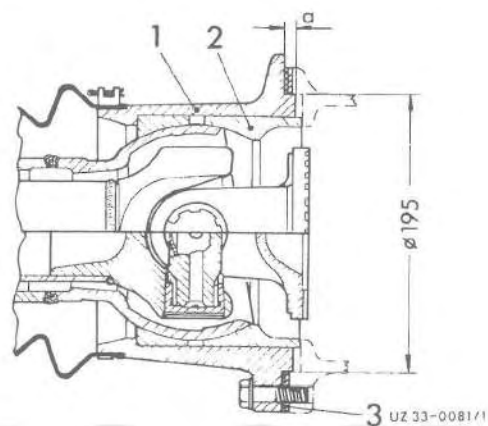


Item	Designation	Modification
1	Torque ball housing	Collar dia. 195 mm shortened from 8 mm to 5 mm (see ill. Item a or b, resp.)
2	Thrust ball bearing shell front	Overall length shortened from 66.5 mm to 62 mm
3	Change in shape	see ill.
4	Sealing compound (Terostat 56)	Used as seal between torque ball housing and transmission flange

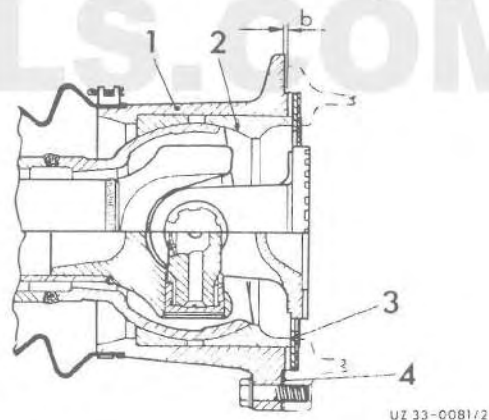
4 Position of shims (3) previously
a = 8 mm

Note:

With installation of SA 35 973 and SA 35 974 (front and rear axle parts with fording ability) the sealing ring previously installed is not required.



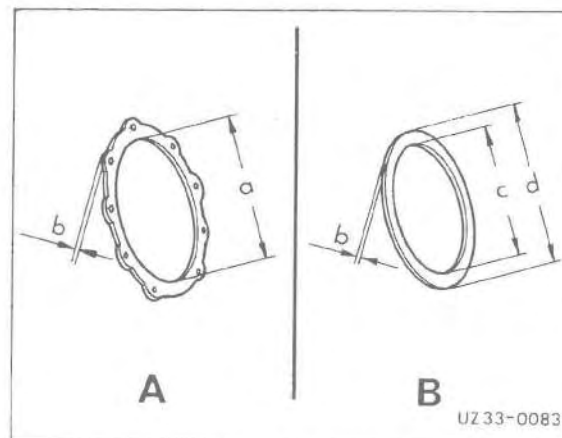
5 Position of shims (3), new



6 Install shims for measuring.

Note:

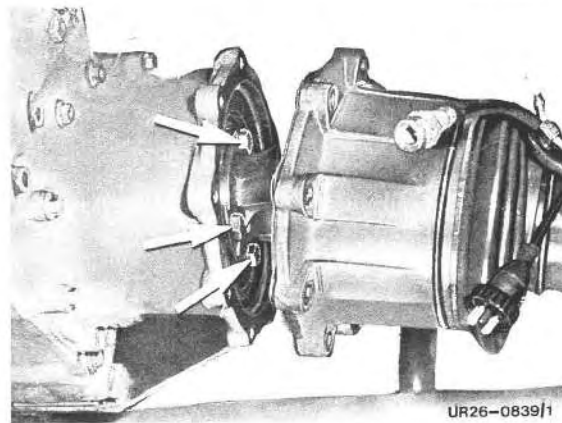
- A Shims, previous
- B Shims, new
- a Dia. 205 mm
- b 0.2; 0.5; 1.0 mm
- c Dia. 148 mm
- d Dia. 194 mm



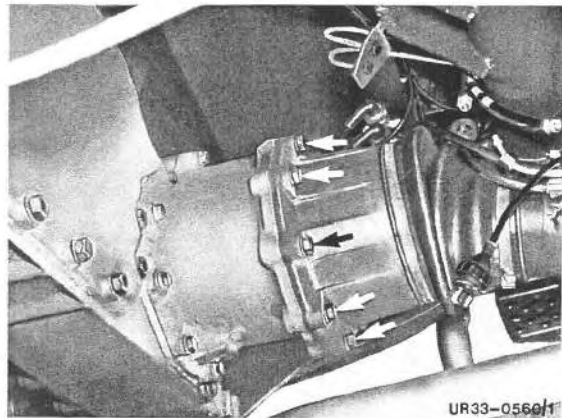
737.111



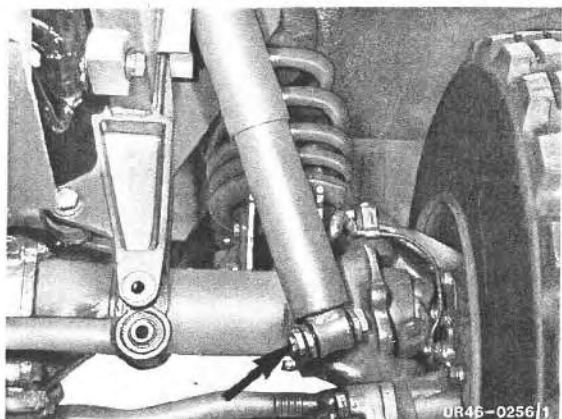
UR33-0553



UR26-0839/1



UR33-0560/1



UR46-0256/1

7 Attach special tool No. 15 and **evenly** tighten fastening bolts until a manual force of 200 to 300 N is achieved at the extension tube.

Note: Remove shims if manual force required is inadequate, and add if excessive.

If using the new shims, shims must be inserted if the manual force required is inadequate. If the manual force required is excessive, remove shims.

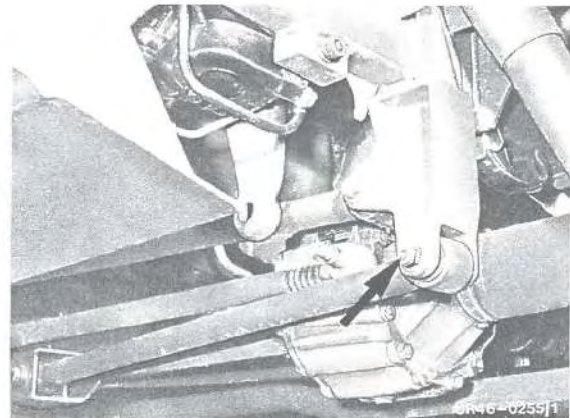
Installation

- 1 Raise vehicle and move axle into position.
- 2 Attach propeller shaft to transmission.

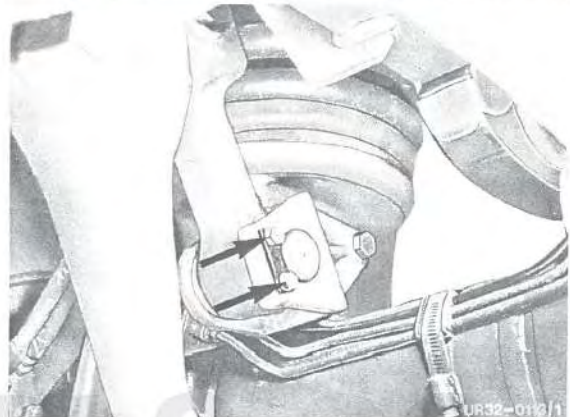
3 Attach and tighten torque ball housing with shims of calculated thickness.
See 1.5/1 for tightening torques.

4 Screw on shock absorber.
See 1.5/1 for tightening torques.

- 5 Bolt transfer link to frame.
See 1.5/1 for tightening torques.

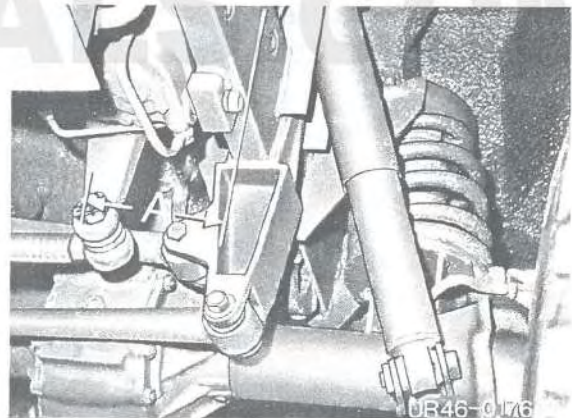


- 6 Install pin at stabilizer and secure.

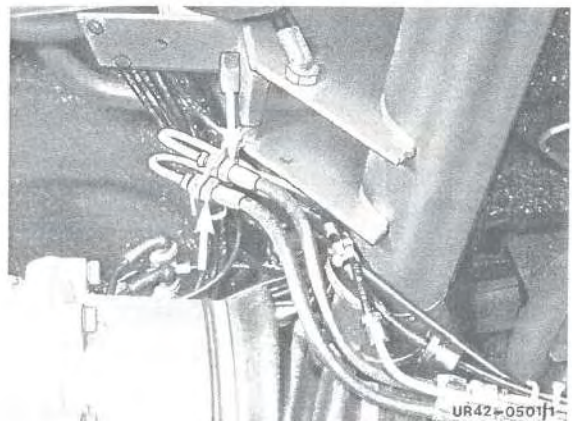


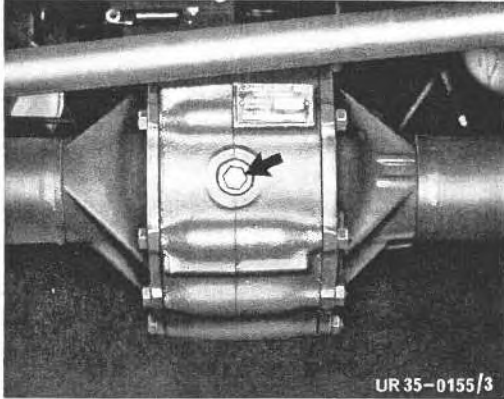
- 7 Insert steering rod in pitman arm, tighten fastening nut and secure.
See 1.5/1 for tightening torques.

Note: Check dim. A from centre of pitman arm to outer edge of frame, adjusting if necessary.
See 1.3/1 for adjusting dim.

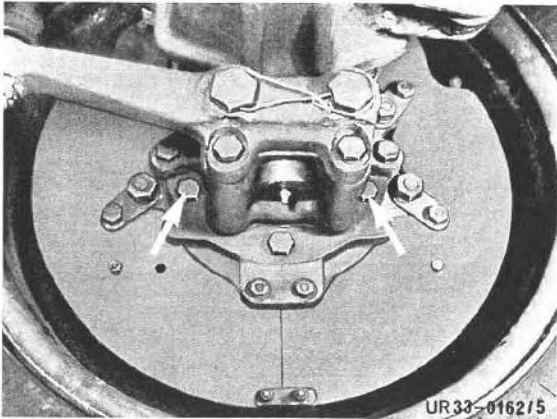


- 8 Fit brake lines for front brake circuits, vent and bleeder lines, line to differential lock and cable for brake shoe wear indicator.





9 Fill center axle housing with oil, referring to 1.1/2 for quantity.



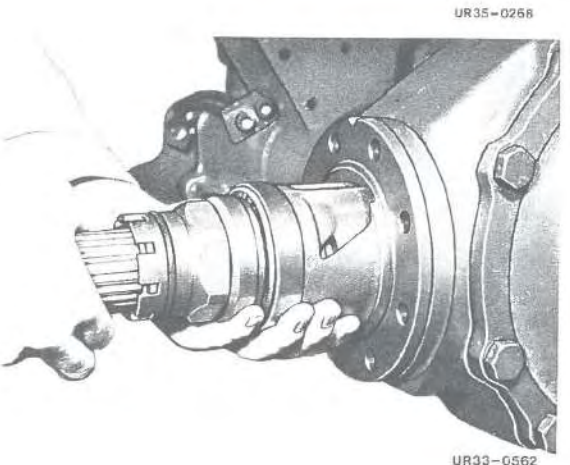
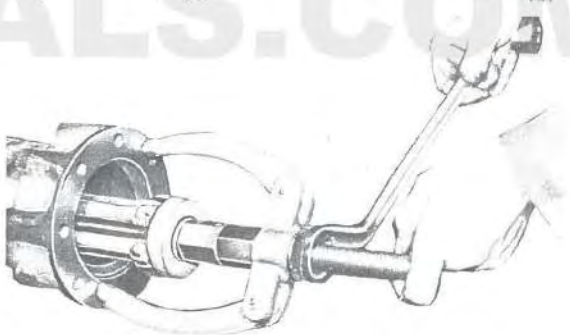
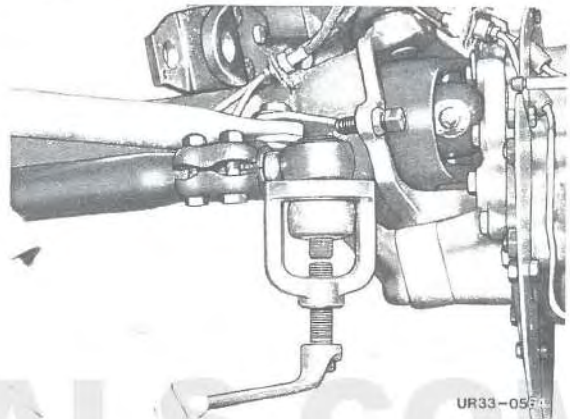
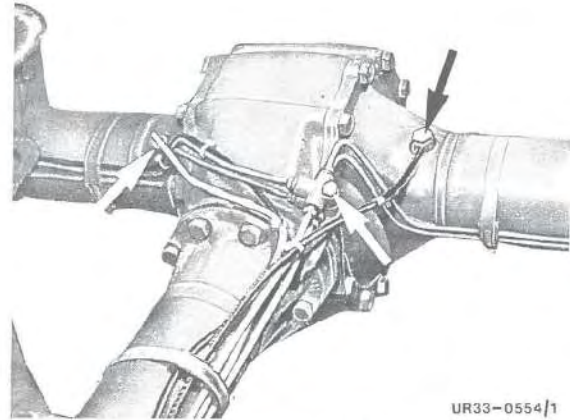
10 Fill wheel hub drive with oil, referring to 1.1/2 for quantity.

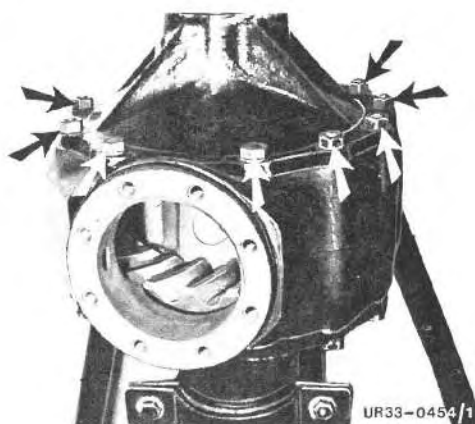
11 Bleed brake system, referring to 42.11-6.1/1.

12 Check brake system for leakage and satisfactory performance.

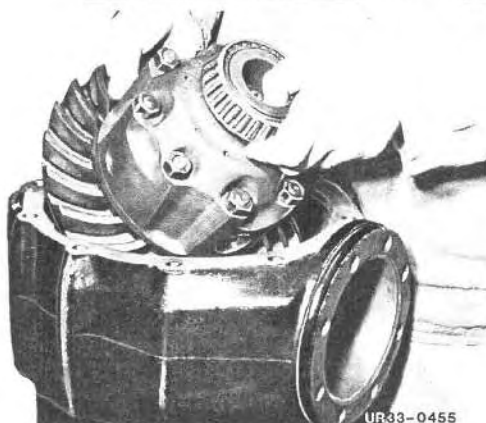
Disassembly

- 1 Remove front axle, referring to 2.1/1.
 - 2 Clean front axle, clamp in trestle and detach wheels.
 - 3 Unscrew brake lines, differential lock line and axle ventilation line.
 - 4 Release and unscrew castle nut, withdrawing drag link and track rod with special tool No. 8.
 - 5 Unscrew axle struts.
 - 6 Unscrew torque tube.
 - 7 Remove propeller shaft.
 - 8 Remove wheel hub drive, referring to 6.1/1.
 - 9 Check shaft seal in torque tube for wear, withdrawing with puller and special tool No. 9 if necessary.
 - 10 Remove drive pinion.
- Note:** Watch out for shims between pinion flange and axle housing.
- 11 Turn front axle by 90° so that long axle tube is pointing upward.





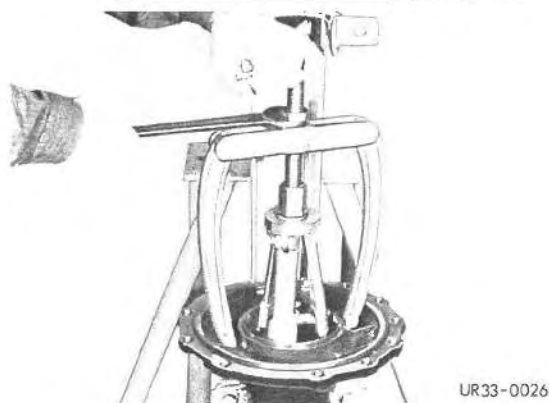
12 Unscrew axle tube at center axle housing.



13 Take differential out of center axle housing.

14 Remove center axle housing.

15 Take release spring, shift dog, shims and piston out of short axle tube.



16 Using puller and special tool No. 9, draw outer races of taper roller bearing out of both axle tubes.

17 Clean and check all parts, exchange if necessary.



Assembly

Note: Prior to assembling axle, be sure to adjust axle drive with reference to „Work Sheet for Adjusting Axle Drive“. During the operation „Adjustment of Axle Drive“ the axle tube at the lock side has already been clamped in trestle, and outer taper roller bearing race and shims have been pressed into position using special tool No. 24.

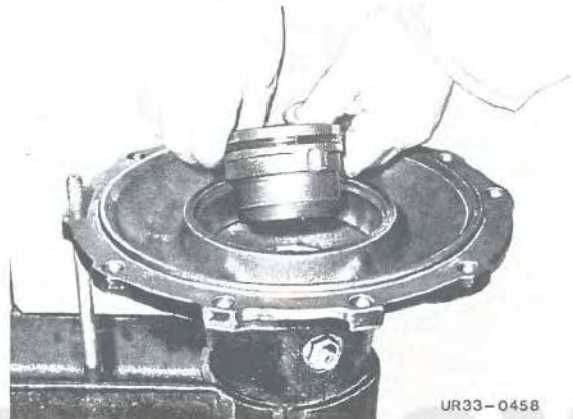
- 1 Using special tool No. 24, press outer taper roller bearing race and shims into long axle tube.



UR33-0457

- 2 Coat piston of differential lock with anti-corrosion oil No. 7 and push sealing rings fully home into short axle tube.

Note: Make sure that adapter socket is seated correctly relative to guide slot in piston.



UR33-0458

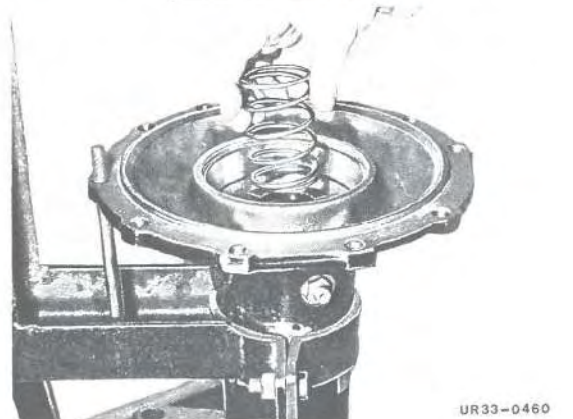
- 3 Fit shift dog with thrust washers and shims in piston.

- 4 Check differential lock for leakage, referring to 7.1/1.



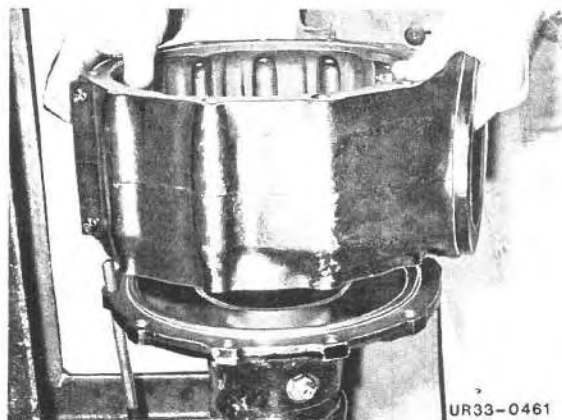
UR33-0459

- 5 Introduce release spring into shift dog.

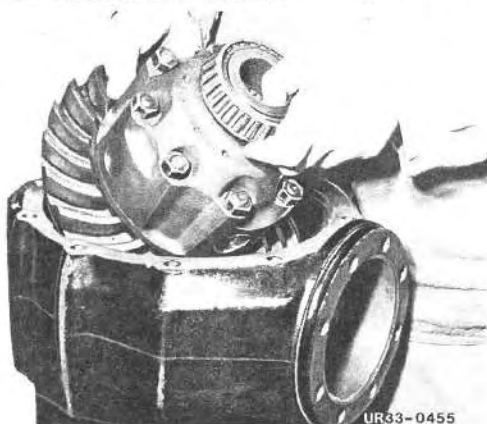


UR33-0460

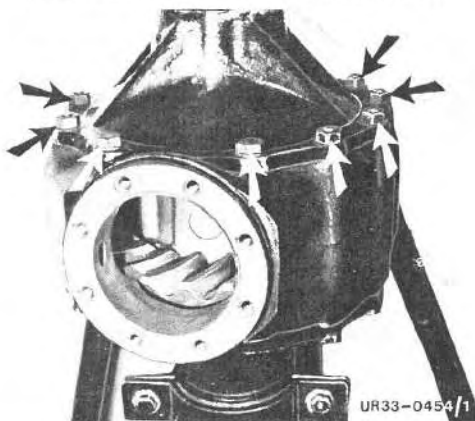
737.111



- 6 Coat contact surface of short axle tube with sealant No. 9. Place center axle housing in position.



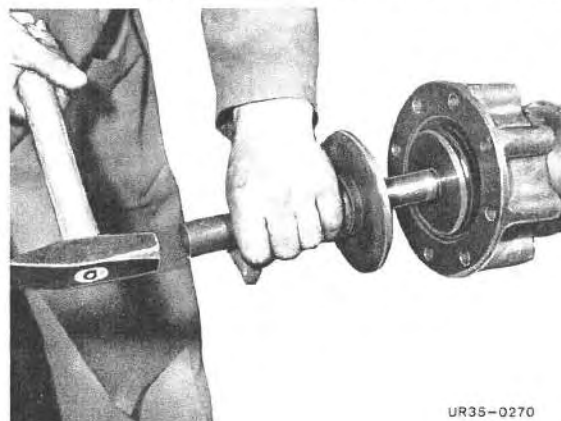
- 7 Fit differential.



- 8 Coat contact surface of long axle tube with sealant No. 9. Torque axle tube to 200 Nm, coating bolts with sealant No. 1. See 1.5/1 for tightening torques.

Note: Secure short bolts with wire.

- 9 Turn front axle by 90° in trestle.



- 10 Coat pinion flange with sealant No. 1 and drive pinion bearing with paste No. 3. Install shims of thickness calculated with reference to work sheet.

Note: Oil passage on pinion flange is to point upward.

- 11 Attach wheel hub drive, referring to 6.1/1.

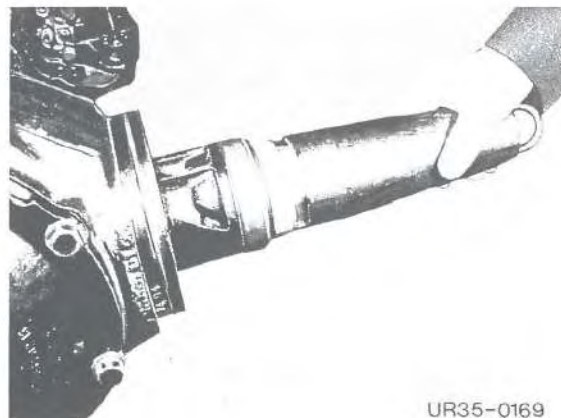
- 12 Coat shaft seal with sealant No. 10 and install in torque tube using special tube No. 11.

737.111

13 Slide special tool No. 16 onto drive pinion, coat torque tube flange with sealant No. 1, bolt and torque. Torque tube to pinion flange. See 1.5/1 for tightening torques.

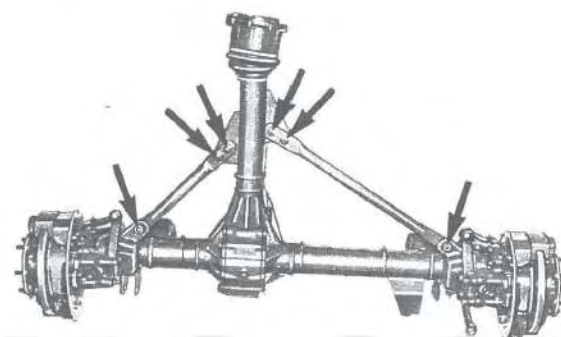
14 Withdraw special tool No. 16 from torque tube.

15 Clean inner splines of propeller shaft and propeller shaft inside, fill cavity behind profil with lubricating grease No. 2 (50 gr.) and install propeller shaft.



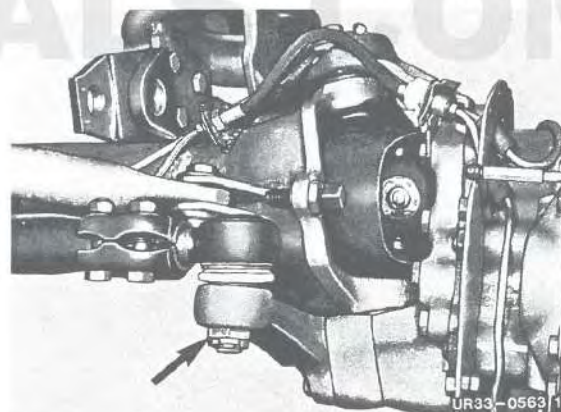
UR35-0169

16 Coat fastening bolts with locking fluid No. 4 and torque axle struts. See 1.5/1 for tightening torques.



UR33-0550/2

17 Attach drag link and track rod. Torque and secure castle nut. See 1.5/1 for tightening torques.



UR33-0563/1

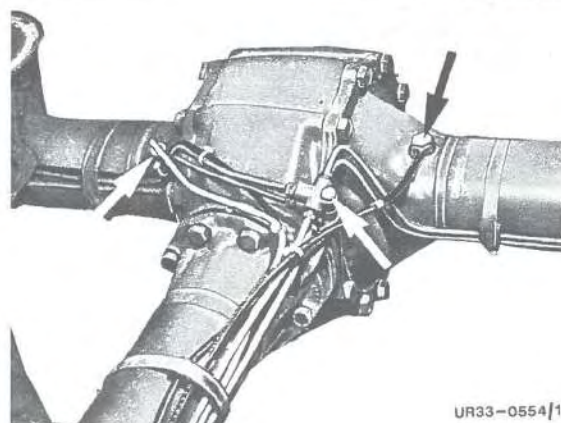
18 Attach brake line, differential lock line and axle ventilation line.

19 Attach wheels and torque wheel nuts. See 1.5/1 for tightening torques.

20 Take front axle off trestle.

21 Install front axle, referring to 2.1/1.

Note: Retorque wheel nuts after 50 km.

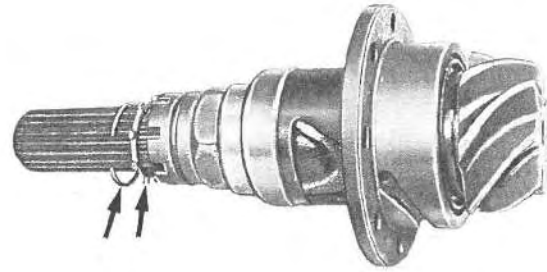


UR33-0554/1

Disassembly

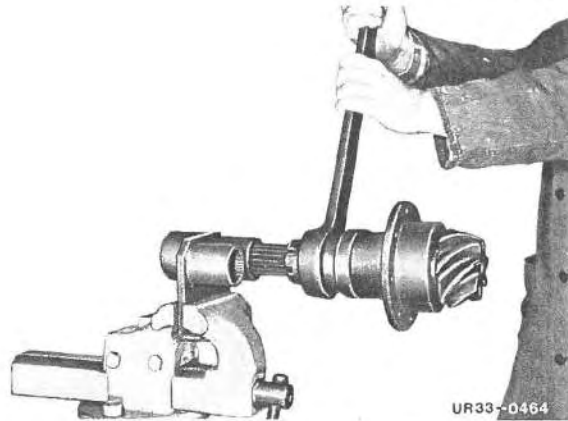
1 Remove pinion flange with drive pinion, referring to 2.1/1.

2 Remove lock washer, serrated washer and sealing ring from hexagon nut.



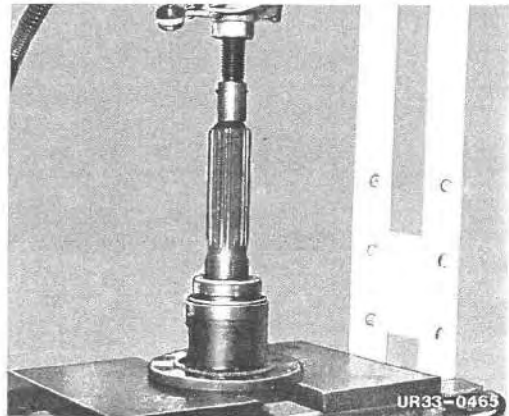
UR33-0463/1

3 Clamp drive pinion with special tool No. 23 and unscrew hexagon nut using special tool No. 17.



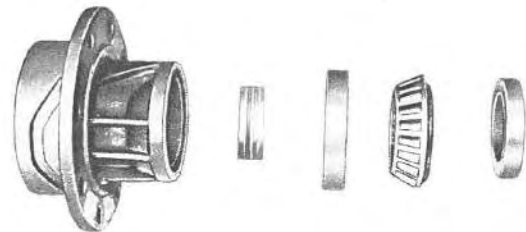
UR33-0464

4 Force drive pinion out.



UR33-0465

5 Remove retaining ring, taper roller bearing ring, bearing ring and clamping sleeve from pinion flange.

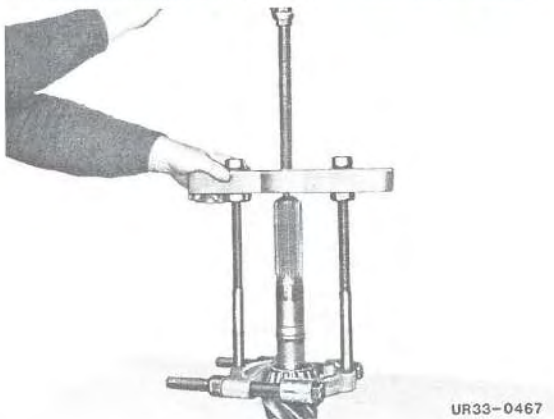


UR35-0221

737.111



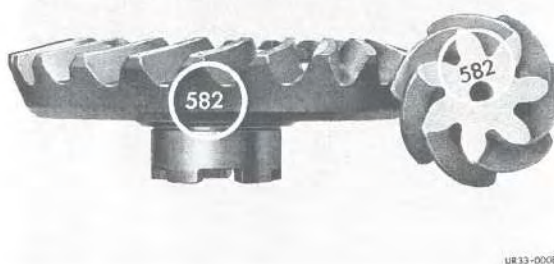
- 6 Withdraw outer taper roller bearing race from pinion flange, using puller and special tool No. 9.



- 7 Pull taper roller bearing and inner race off drive pinion.

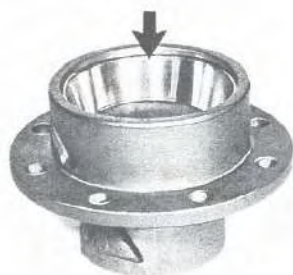
- 8 Clean and check all parts, exchange if necessary.

Note: Prior to installing taper roller bearing, rinse out and insert with SAE 90 oil.



Important:

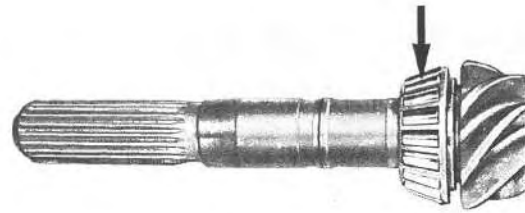
If drive pinion is to be exchanged, it must always be replaced **together** with ring gear. Make sure serial numbers are identical.



Assembly

- 1 Force outer taper roller bearing race into pinion flange.

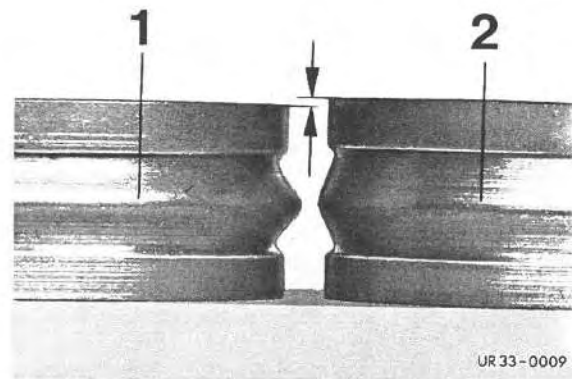
- 2 Press taper roller bearing onto drive pinion.
- 3 Introduce drive pinion into pinion flange.



UR33-0468/1

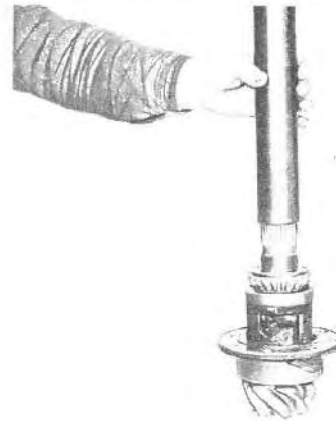
- 4 Slip new clamping sleeve onto drive pinion.

Note: Be sure to use new clamping sleeve (2) because old sleeve (1) is compressed from first use.



UR33-0009

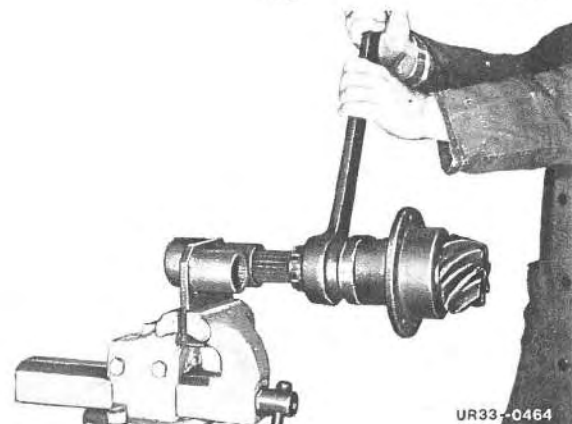
- 5 Coat taper roller bearing with lubricating grease No. 2. Press taper roller bearing into position so that it has 1 to 2 mm clearance.



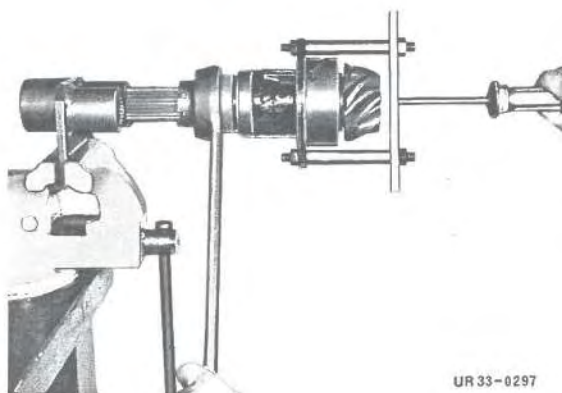
UR33-0495

- 6 Fit retaining ring and screw hexagon nut on.

- 7 Clamp drive pinion with special tool No. 23 and torque hexagon nut with special tool No. 17 until bearing play disappears.



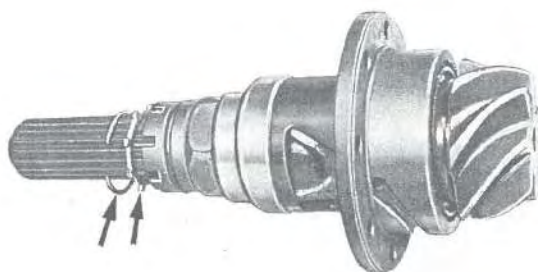
UR33-0464



UR 33-0297

8 Apply special tools No. 20 and No. 21 to pinion flange and tighten hexagon nut with special tool No. 17 until break-away torque of 3 to 3.5 Nm is reached.

Note: Break-away torque is reached just before hexagon nut leaves neutral position. Drive pinion will have to be removed and operations 3 to 8 repeated if torque setting is exceeded.



UR33-0463/1

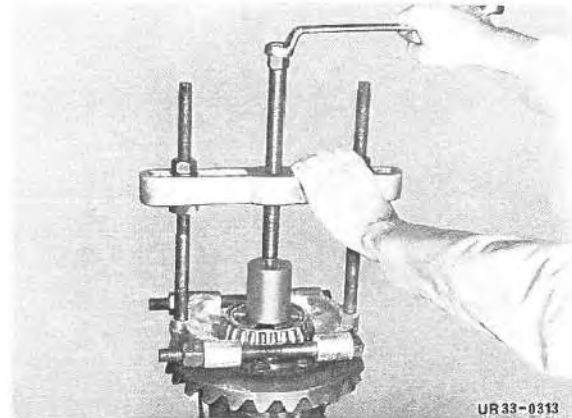
9 Fit serrated washer, lock washer and sealing ring in hexagon nut.

10 Coat drive pinion with paste No. 3 and install with pinion flange.

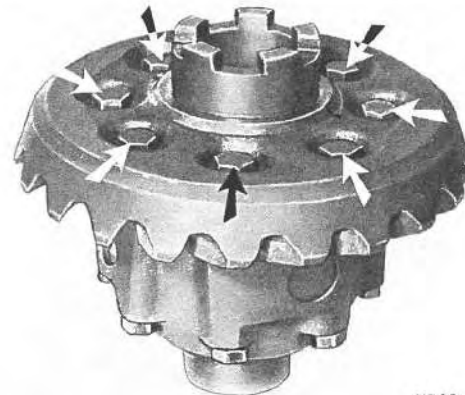
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Disassembly

- 1 Remove front axle, referring to 2.1/1.
- 2 Disassemble front axle, referring to 3.1/1.
- 3 Pull both taper roller bearings off differential.



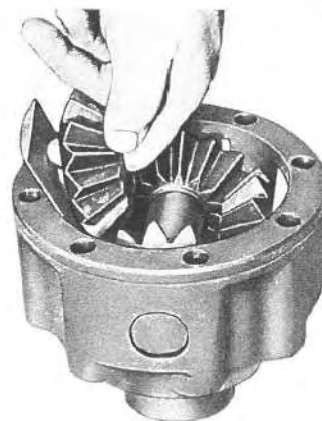
- 4 Unscrew and remove ring gear from differential housing.

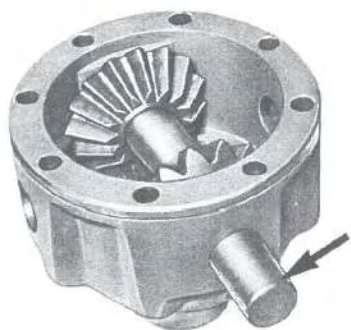


- 5 Remove side gear and thrust washer.
- 6 Remove both short differential pinion shafts.

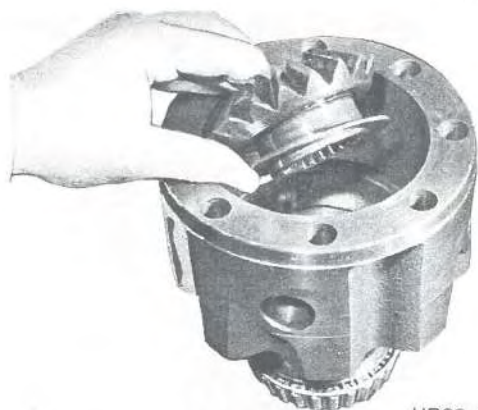


- 7 Take thrust segments and differential pinion shafts out.





UR33-0472/1



UR33-0330

- 8 Remove long differential pinion shaft.
- 9 Take thrust segments and differential pinions out.

10 Take side gear and thrust washer out of differential housing.

11 Clean and check all parts, exchange if necessary.

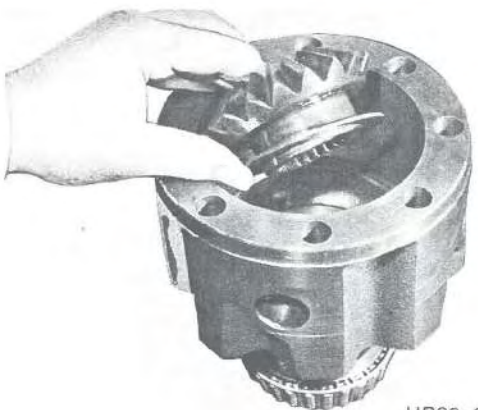


UR33-0008

Assembly

Note: Ring gear must always be exchanged together with drive pinion. Make sure serial numbers are identical.

1 Coat all differential pinion shafts and thrust segments with paste No. 3.



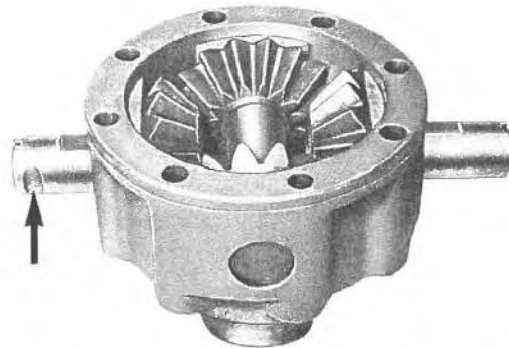
UR33-0330

2 Fit thrust washer and side gear in differential housing.

- 3 Fit thrust segments, differential pinions and differential pinion shafts.

Note: Note recesses in differential pinion shafts for bolts.

- 4 Insert thrust washer and side gear in differential housing.



UR33-0473/1

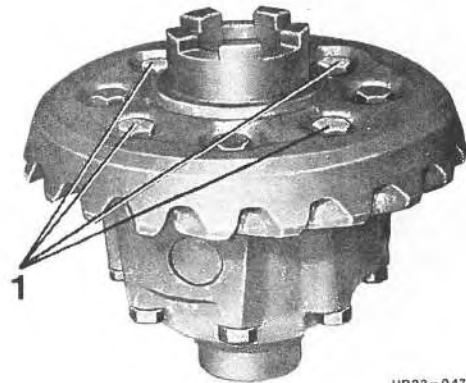
- 5 Place ring gear on differential housing and torque fastening bolts.
See 1.5/1 for tightening torques.

Note:

From Model Chassis end No. Front axle No.
435.115 001 522

Body-fit bores (1) with colour dot (white) are marked and reamed with reamer 13 H 7, install body-fit bolts only here.

Ream holes of differential housing without marking (1) with reamer 13 H 7 as before.



UR33-0474/1

- 6 Using special tool No. 25, press large inner taper roller bearing race onto differential housing and coat with fluid grease No. 6.



UR33-0475

- 7 Using special tool No. 25, press small inner taper roller bearing race onto differential housing and coat with fluid grease No. 6.

- 8 Assemble front axle, referring to 3.1/1.

- 9 Install front axle, referring to 2.1/1.



UR33-0476

Disassembly

Note: Operations at left and righthand wheel hub drives are to be executed in same order. Work is same for axle in situ.

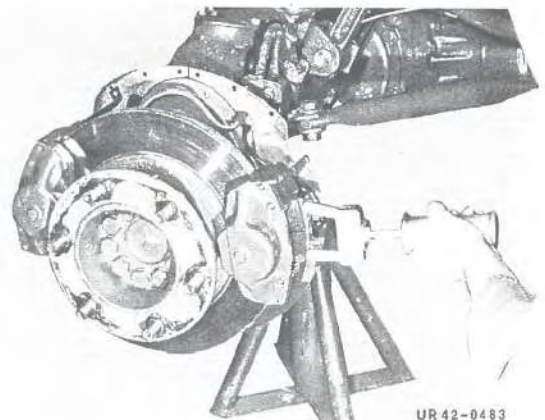
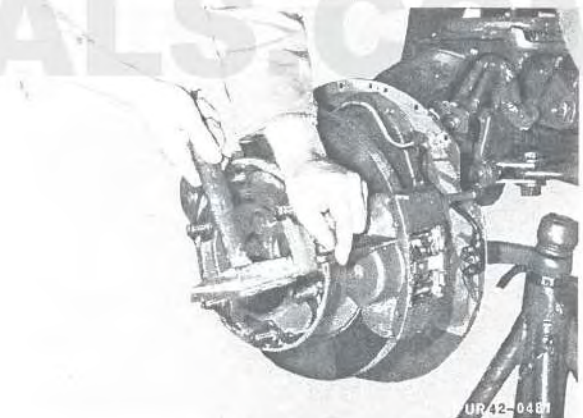
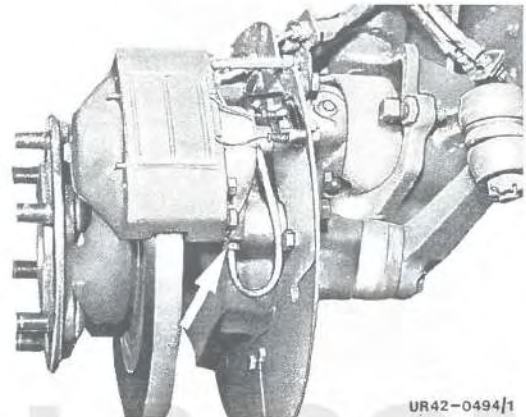
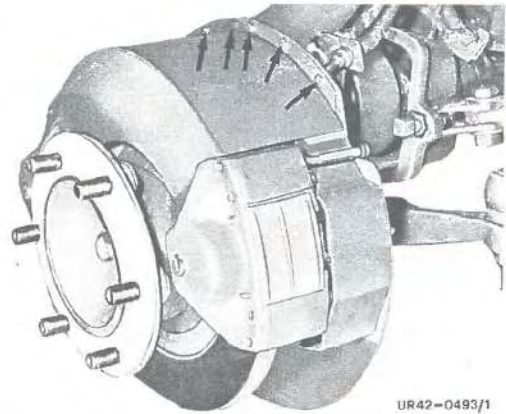
- 1 Remove front axle, referring to 2.1/1.
- 2 Clean front axle, clamp in trestle and detach wheels.
- 3 Unscrew brake backplate.
- 4 Remove guard plates.
- 5 Unscrew brake lines and seal ends.

Note: With axle in situ, brake lines can be left attached to fixed caliper.

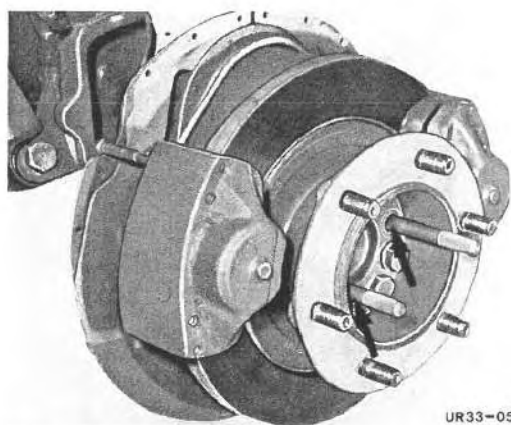
- 6 Remove metal cover from fixed caliper, knock pins out and detach spring clip.

Note: Detach brake lining wear indicator from left-hand fixed caliper at rear.

- 7 Remove brake pads using special tool No. 10.
- 8 Remove shields.



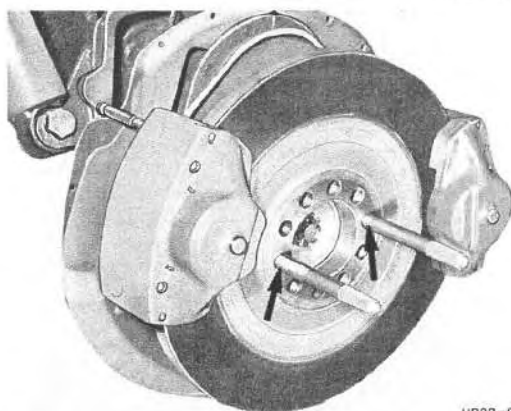
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UR33-0555/1

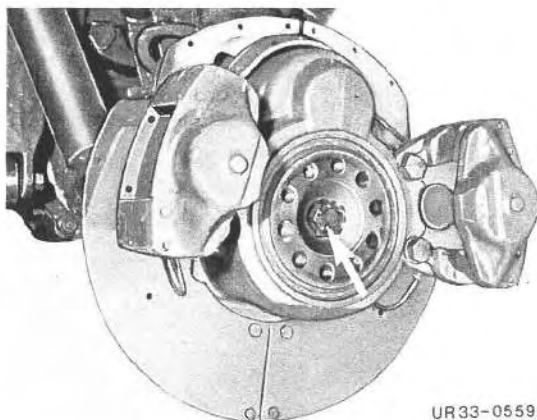
9 Unscrew two wheel hub fastening bolts using special tool No. 3

10 Insert special tool No. 4 and unscrew all other fastening bolts. Remove wheel hub.



UR33-0556/1

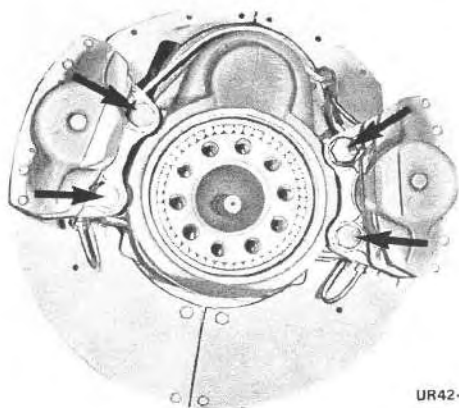
11 Unscrew special tool No. 4 and remove with brake disk.



UR33-0559/1

12 Release and unscrew castle nut on clamping bolt.

13 Remove wear ring.

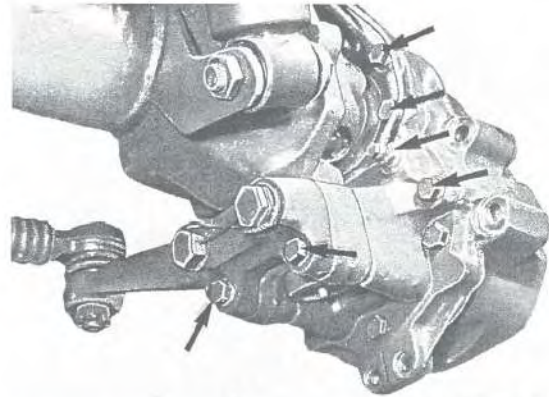


UR42-0500/1

14 Unscrew both fixed calipers with metal covers.

Note: Unscrew wear indicator plug at metal cover.

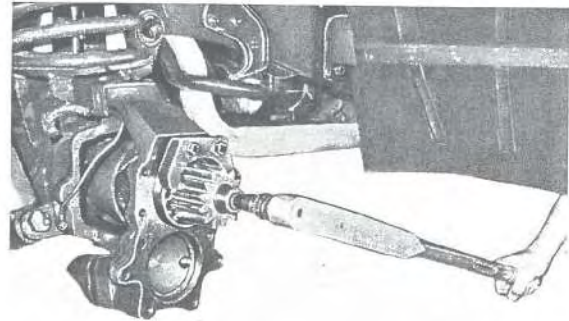
- 15 Unscrew bleeder line and brake line holder, detaching wheel hub drive.



UR33-0546/1

- 16 Lock drive gear using special tool No. 6 and unscrew center fastening bolt using special tool No. 2.

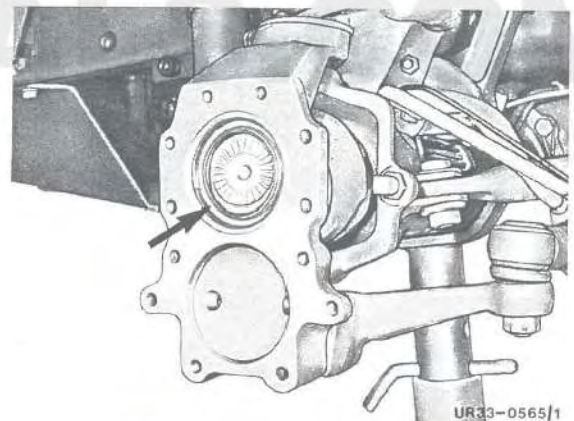
- 17 Remove drive gear with roller bearing.



UR33-0544

Removal and installation of sealing ring, steering knuckle attached.

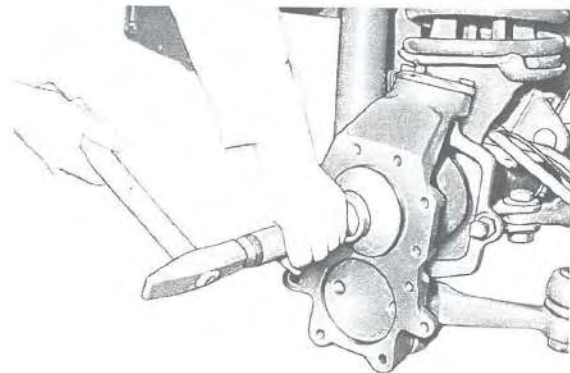
- 18 With front axle in situ, turn steering wheel to lock and remove shaft seal.



UR33-0565/1

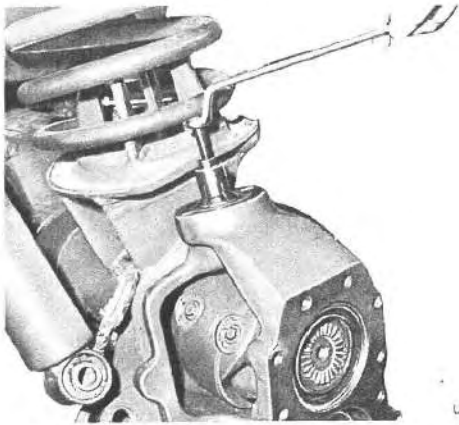
- 19 Coat shaft seal with lubricating grease No. 2 and install using special tool No. 12.

- 20 Install wheel hub drive, referring to 6.1/9 to 6.1/11, items 19 to 32.



UR33-0566

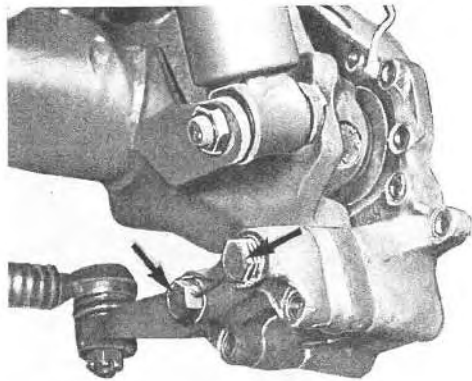
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UR33-0543

- 21 Unscrew end plate, removing upper swivel joint with special tool No. 7

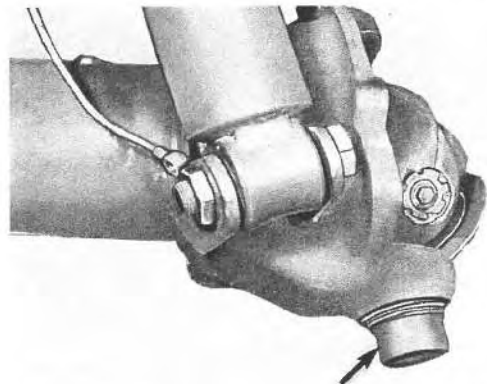
Note: With axle in situ unscrew shock absorber at axle and raise vehicle.



UR33-0547/1

- 22 Release and unscrew fastening bolts. Remove steering knuckle arm and steering arm with shell bearing half, thrust piece, thrust shell, sliding lock and needle bearing.

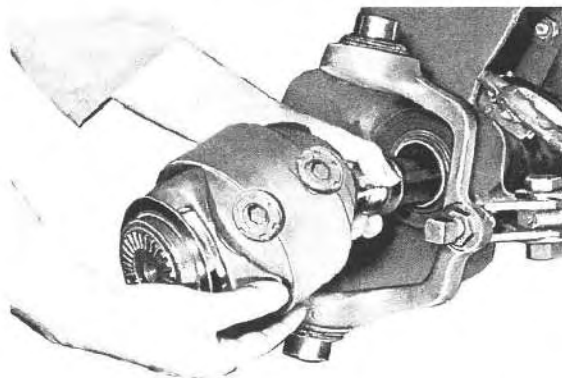
Note: These arms are only to be unscrewed after axle has been raised, because fastening bolts hold not only lower swivel joint but also bearing cap.



UR33-0548/1

- 23 Remove steering knuckle.

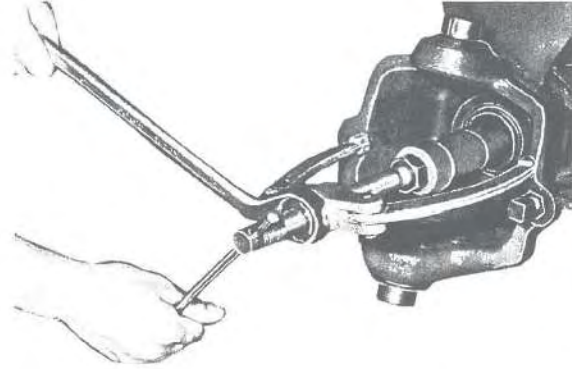
- 24 Remove bearing bush at bottom with shim and shaft seal.



UR33-0549

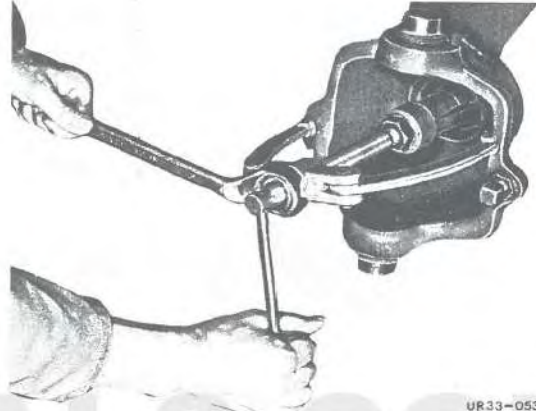
- 25 Remove constant-velocity joint shaft.

26 Check both shaft seals in the axle tube for wear, remove as required using puller and counter-support No. 9.



UR33-0537

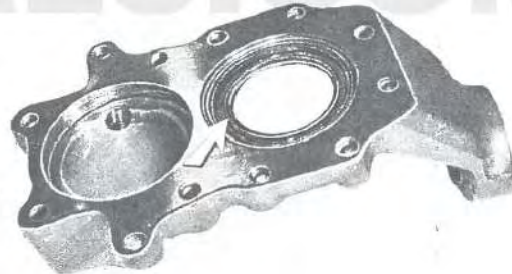
27 Remove circlip and withdraw needle bearing with puller and counter-support No. 9. Do not damage circlip behind the bearing



UR33-0539

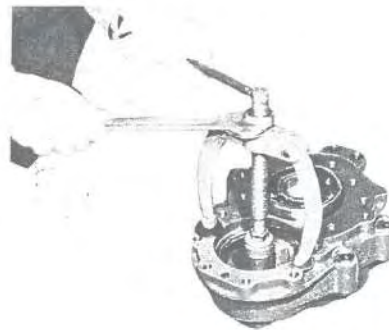
Removing Sealing Ring, Steering Knuckle Removed

28 Remove shaft sealing ring from steering knuckle.



UR33-0535/1

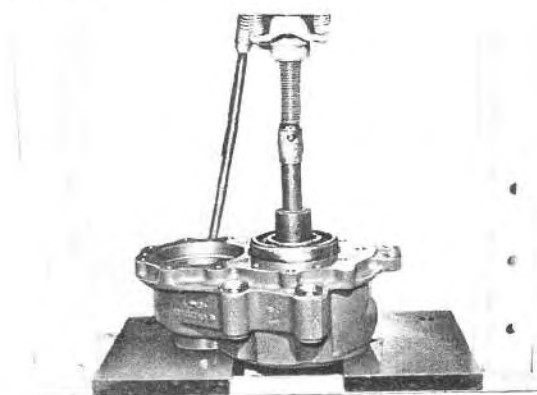
29 Check small roller bearing in wheel hub drive housing for wear and remove if necessary.



UR35-0266

33.6 Disassembly and Assembly of Wheel Hub Drive

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UR35-0265

30 Press drive gear together with roller bearing and shaft sealing ring out of wheel hub drive housing.

Note: Slip assembly ring onto roller bearing so that rollers cannot fall out.

31 Remove bearing in wheel hub drive housing.

32 Clean all parts, check and renew if necessary.

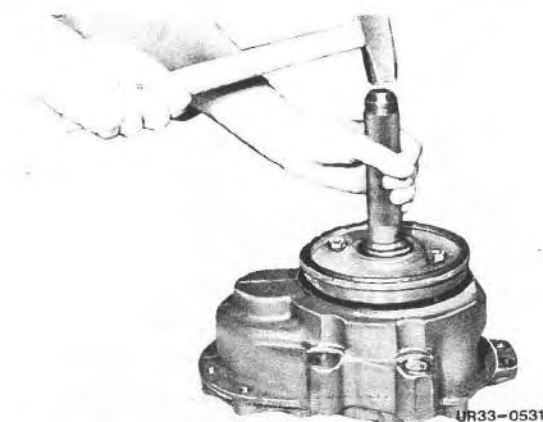


UR35-0264

Assembly

Note: Prior to assembly determine difference between bearing seat in the steering knuckle and wheel hub drive housing as well as the thickness of the grooved ball bearing and pack with shim rings.

1 If small roller bearing in the wheel hub drive housing is defective, fit new bearing using special tool No. 26 and coat with fluid grease No. 6.



UR33-0531

Note: for wheel hub drive housing part No. 416 331 0405 the bearing seat must be balanced with 1 shim 0,3 mm and 5 shims each 0,5 mm.

2 Install driven gear in wheel hub drive housing and press in deep-groove ball bearing. Fit sealing ring on deep-groove ball bearing.

3 Press cylindrical roller bearing into wheel hub drive housing using special tool No. 13.

Note: Remove assembly ring from cylindrical roller bearing only when pressing in the bearing.



UR33-0534

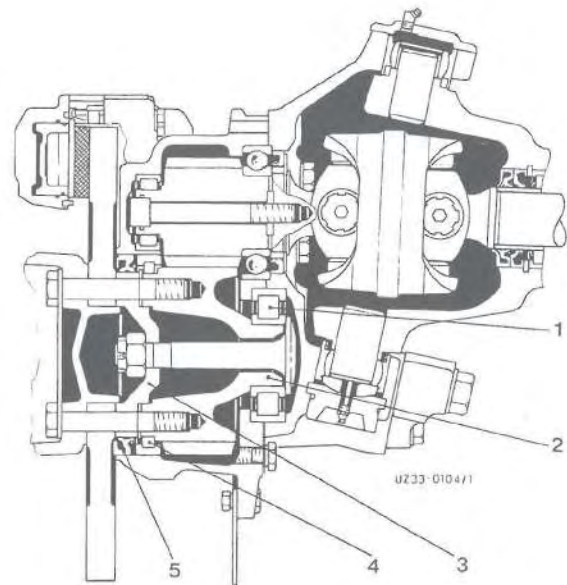
4 Coat shaft sealing ring with lubricating grease No. 2, press onto wear ring and install using special tool No. 13.

5 Fit clamping bolt and screw on castle nut.

Note: The following parts are exchanged in the (SA 36 183) area of the driven gear during conversion work to reinforced wheel bearing:

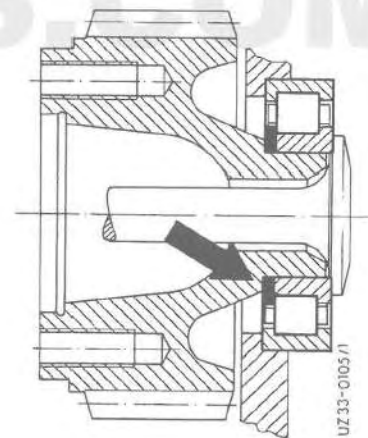
New Parts

- 1 Cylindrical roller bearing
- 2 Drive gear
- 3 Wear ring
- 4 Cylindrical roller bearing
- 5 Shaft sealing ring, (previous version)



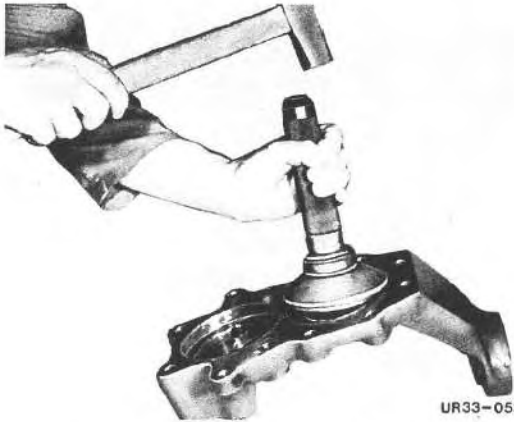
Care must be taken during installation of the inner cylindrical roller bearing to ensure that the thrust washer of the inner bearing race is fitted facing the gear side. This ensures that during subsequent removal, the bearing is not removed in the steering knuckle, but rather together with the driven gear, thereby ensuring correct assembly.

Assemble the reinforced wheel bearing as described on page 6.1/6.

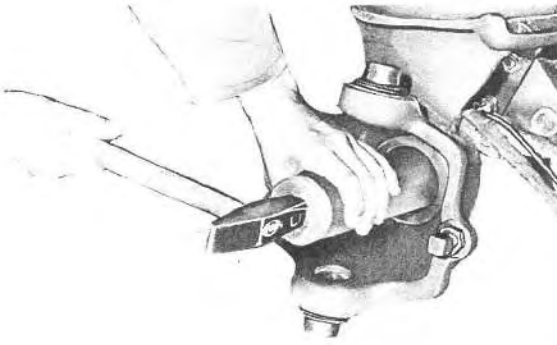


33.6 Disassembly and Assembly of Wheel Hub Drive

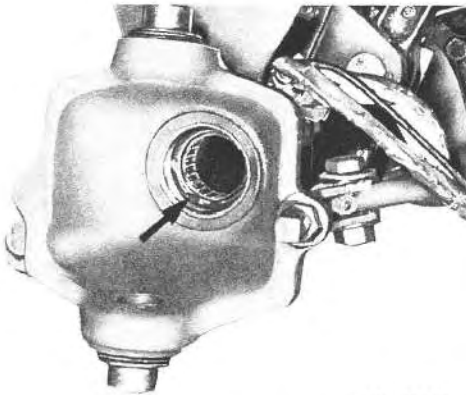
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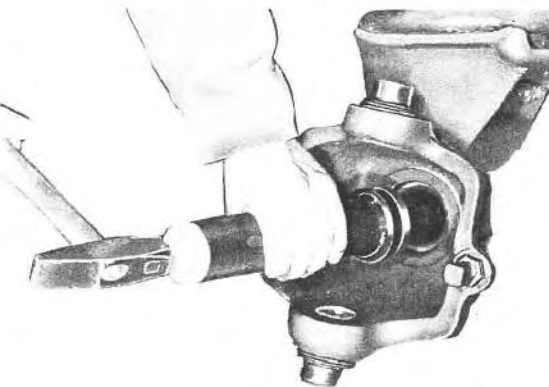
6 Coat shaft sealing ring with lubricating grease No. 2 and press into the steering knuckle using special tool No. 12.



7 Using special tool No. 19 press needle bearing into axle tube up to the rear circlip.

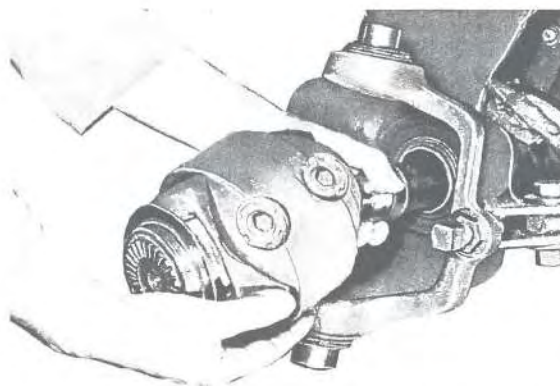


8 Install front circlip.



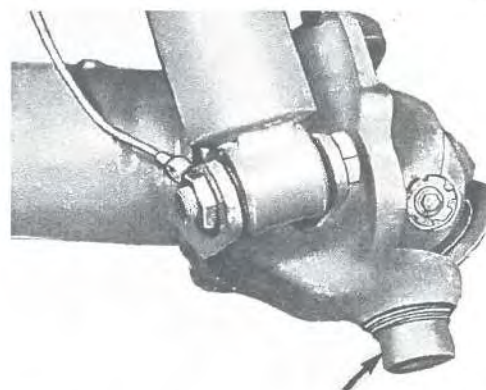
9 Coat outside of both shaft sealing rings with sealing compound No. 5 and press in using special tool No. 18.

10 Coat spline on constant-velocity joint shaft with running - in paste No. 3 and insert shaft. Coat inner diameter of wear ring with sealing compound No. 5.



UR33-0549

11 Coat shaft sealing ring with lubricating grease No. 2 and install with shim and bearing bushing.

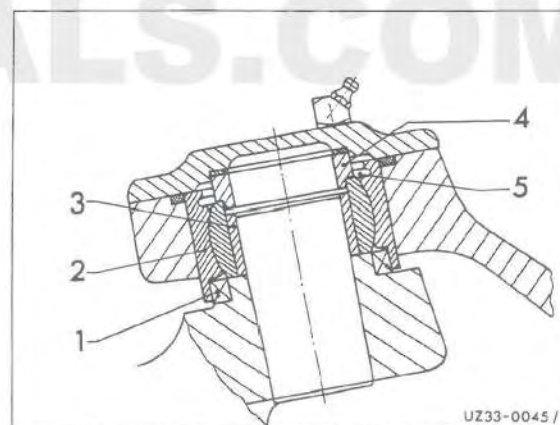


UR33-0548/1

12 Coat shaft sealing ring (1) with lubricating grease and press into upper plain bearing. Coat plain bearings (2) with running - in paste No. 3 and press onto kingpin with bushing (3).

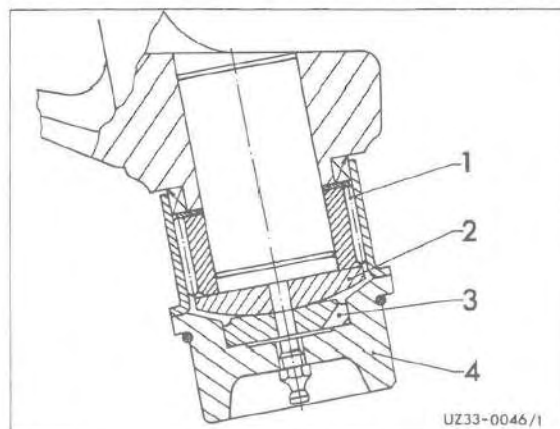
Note: To assist removal of plain bearing (2), install extractor groove (5) of the plain bearing facing upwards.

13 Attach preassembled steering knuckle to kingpin.

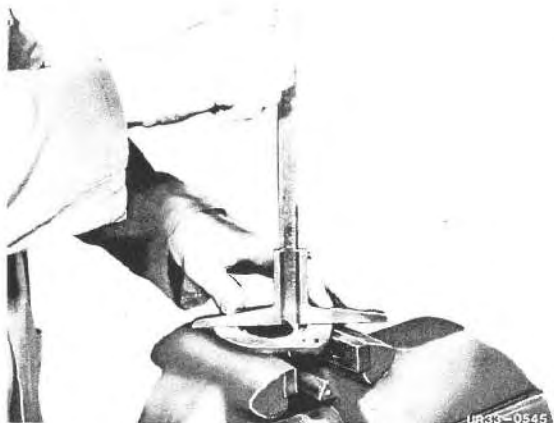


UZ33-0045 /1

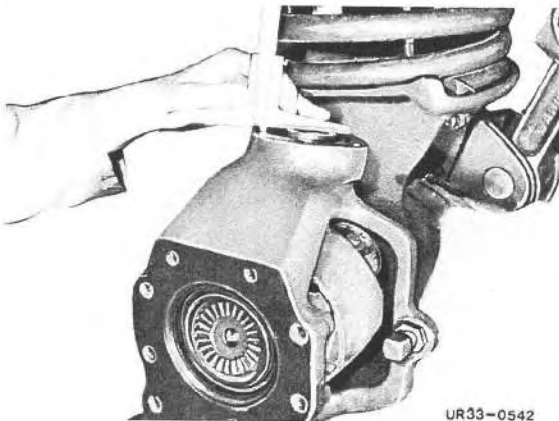
14 Coat half bearing shell and fastening bolts with sealing compound No. 1. Tighten needle bearing (1), thrust piece (2), fulcrum pad (3), thrust shell (4) and half bearing shell with steering knuckle arm and secure. For tightening torque, refer to 1.5/1.



UZ33-0046 /1



15 Determine depth for installation of spacer ring (4) in the end plate.

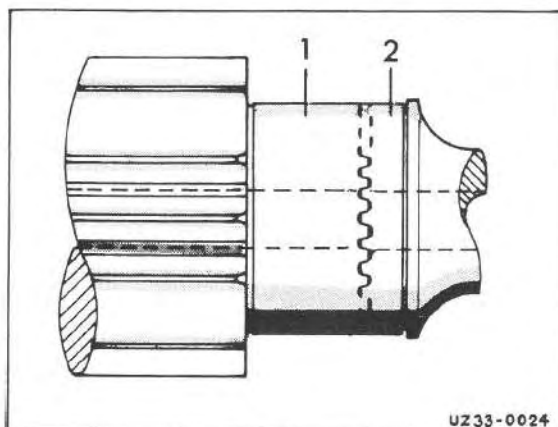


16 Fit spacer ring (4) in plain bearing, determine distance between spacer ring and contact surface of steering knuckle.

17 The difference between the two measurements plus permissible preload must be packed with shim rings. For permissible preload, refer to 1.3/1. Measure only when no load is applied to the axle.

Note: Support wheel hub drive if measured when installed.

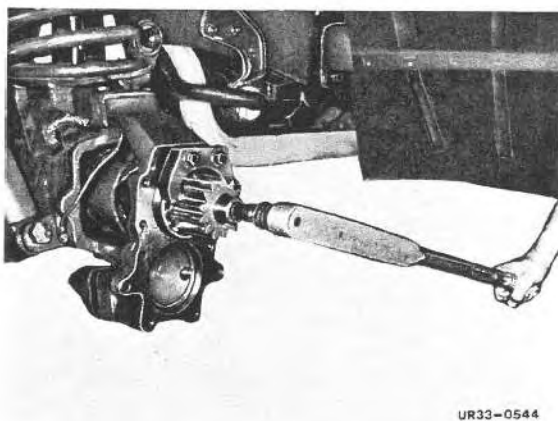
18 Insert sealing ring and bolt end plate to knuckle.



19 Install drive gear with roller bearing.

Note: Screw in center fastening bolt while turning the drive gear until a backlash is felt.

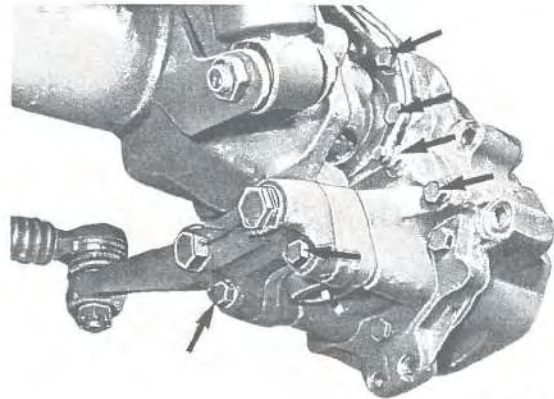
20 Turn fastening bolt by 2 to 2 1/2 more turns and bolt drive gear (1) to constant-velocity joint shaft (2) so that teeth mesh.



21 Block drive gear using special tool No. 6. Tighten center fastening bolt using special tool No. 2. For tightening torques, refer to 1.5/1.

22 Coat contact surface of wheel hub drive housing and fastening bolts with sealing compound No. 1. Attach wheel hub drive housing, breather line and brake line brackets.

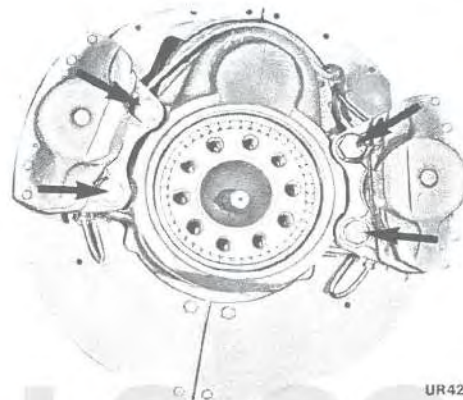
Note: Position rollers of the small roller bearing with multipurpose grease and center wheel hub drive on bearing seats with the aid of bolts, then tighten all remaining fastening bolts.



UR33-0546/1

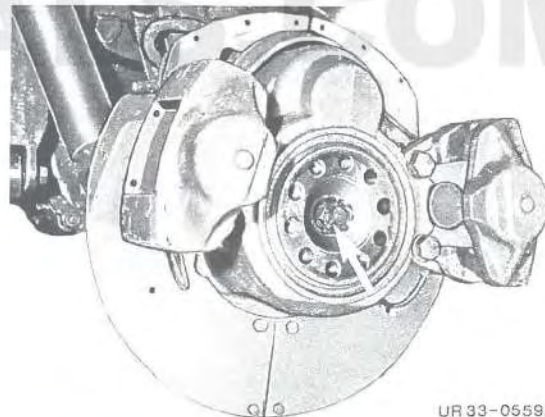
23 Bolt on both fixed calipers with cover plates. Tighten bolts holding fixed calipers. For tightening torque, refer to 1.5/1. Fit close-tolerance bolts for fixed calipers at side where disk brake emerges from caliper (in direction of travel).

24 Screw on cable connector for wear indicator to cover plate.



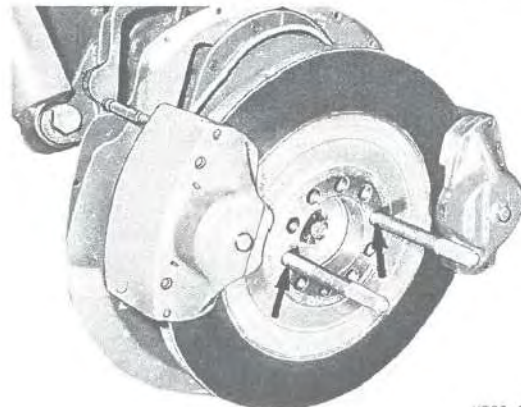
UR42-0500/1

25 Block constant-velocity joint shaft and tighten castle nut of clamping bolt and secure. For tightening torques, refer to 1.5/1.



UR 33-0559/1

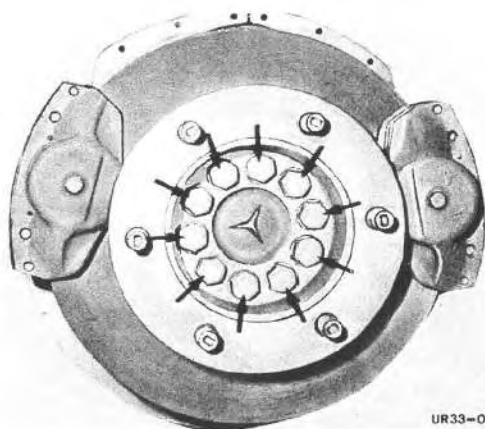
26 Attach brake disk using special tool No. 4.



UR33-0556/1

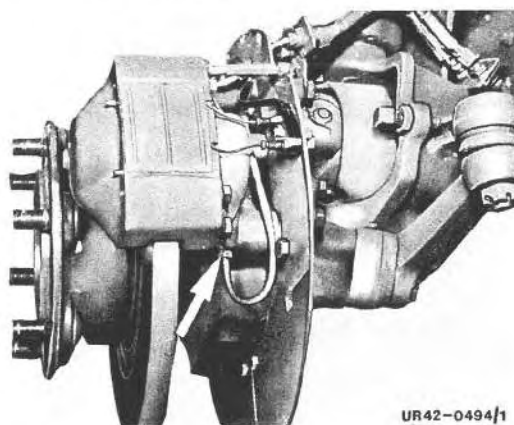
33.6 Disassembly and Assembly of Wheel Hub Drive

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27 Block constant velocity joint shaft, attach wheel hub and tighten fastening bolts using special tool No. 3.

For tightening torques, refer to 1.5/1.



28 Install brake pads.

29 Attach guards.

30 Connect brake line.

31 Attach brake cover plate.

32 Fit road wheels and tighten wheel nuts.
For tightening torques, refer to 1.5/1.

Note: Retighten wheel nuts after 50 km.

33 Install front axle, refer to 2.1/1.

34 Bleed brake system and check for leaks.